THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS
MATH3603 Probability Theory

Assignment 5
Outline of solution

1. Label all the states starting from 1.

   For P_1, we have $P^2_{1,1} > 0 (1 \to 2 \to 1)$ and $P^3_{1,1} > 0 (1 \to 2 \to 3 \to 1)$, thus the gcd of $\left\{n \geq 1 : P^n_{1,1} > 0\right\}$ must be 1. Hence the period of state 1 is 1. Since P_1 defines a Markov chain with only one class, all its states have period 1.

   For P_2, starting from state 3, there are only two possible choices for the first three steps: either $3 \to 1 \to 4 \to 3$ or $3 \to 2 \to 4 \to 3$. Thus we must have n three steps: either $3 \to 1 \to 4 \to 3$ or $3 \to 2 \to 4 \to 3$. Thus we must have n there is only one class for P_2, all states have period 3.

   For P_3, each of the recurrent state in $\{1, 3, 4, 5\}$ has a positive probability to reenter itself after one step. Thus they all have period 1.

2. (a) We have the transition probability matrix

$$P = \begin{bmatrix} 1-3\alpha & \alpha & \alpha & \alpha \\ \alpha & 1-3\alpha & \alpha & \alpha \\ \alpha & \alpha & 1-3\alpha & \alpha \\ \alpha & \alpha & \alpha & 1-3\alpha \end{bmatrix} = (1-4\alpha) I_4 + \alpha J_4.$$

   where I_4 is the 4×4 identity matrix and J_4 is the 4×4 matrix that every entry is 1. Note that $(J_4)^n = 4^{n-1} J_4$ by simple calculation. Therefore

$$\begin{aligned} P^n &= ((1-4\alpha) I_4 + \alpha J_4)^n \\ &= (1-4\alpha)^n I_4 + \sum_{k=1}^{n} \mathrm{C}^n_k (1-4\alpha)^{n-k} \alpha^k \, (J_4)^k \\ &= (1-4\alpha)^n I_4 + \left(\sum_{k=1}^{n} \mathrm{C}^n_k (1-4\alpha)^{n-k} \alpha^k 4^{k-1} \right) J_4 \\ &= (1-4\alpha)^n I_4 + \frac{1}{4} \left(\sum_{k=1}^{n} \mathrm{C}^n_k (1-4\alpha)^{n-k} (4\alpha)^k \right) J_4 \end{aligned}$$

Thus

$$\begin{aligned} p_{ii}^n &= (1-4\alpha)^n + \frac{1}{4}\left(\sum_{k=1}^{n} \mathrm{C}_k^n (1-4\alpha)^{n-k}(4\alpha)^k\right) \\ &= \frac{3}{4}(1-4\alpha)^n + \frac{1}{4}\left(\sum_{k=0}^{n} \mathrm{C}_k^n (1-4\alpha)^{n-k}(4\alpha)^k\right) \\ &= \frac{3}{4}(1-4\alpha)^n + \frac{1}{4}(1-4\alpha+4\alpha)^n \\ &= \frac{1}{4} + \frac{3}{4}(1-4\alpha)^n. \end{aligned}$$

So $p_{11}^n = \frac{1}{4} + \frac{3}{4}(1-4\alpha)^n$.

(b) As $n \to \infty, p_{ii}^n \to \frac{1}{4}$. So the long run proportion of time the chain in each state is $\frac{1}{4}$.

3. Let State 1 be "red ball being drawn", State 2 be "white ball being drawn" and State 3 be "blue ball being drawn". We have the transition probability matrix

$$P = \begin{bmatrix} 1/5 & 0 & 4/5 \\ 2/7 & 3/7 & 2/7 \\ 1/3 & 4/9 & 2/9 \end{bmatrix}$$

Let π_i be the limiting probability. Then we have

$$\begin{bmatrix} -4/5 & 2/7 & 1/3 \\ 0 & -4/7 & 4/9 \\ 4/5 & 2/7 & -7/9 \end{bmatrix} \begin{bmatrix} \pi_1 \\ \pi_2 \\ \pi_3 \end{bmatrix} = \mathbf{0} \Longrightarrow \begin{bmatrix} \pi_1 \\ \pi_2 \\ \pi_3 \end{bmatrix} = \begin{bmatrix} 25s \\ 28s \\ 36s \end{bmatrix}$$

for any $s \in \mathbb{R}$. Since $\sum \pi_i = 1$, we have $\pi_1 = \frac{25}{89}, \pi_2 = \frac{28}{89}$ and $\pi_3 = \frac{36}{89}$.

4. Denote X_n : the number of pairs of shoes at the door the runner starts from at the beginning of day n. Then X_n is a Markov chain with transition probabilities

$$\begin{aligned} P_{i,i} &= \frac{1}{4} \quad 0 < i < k \\ P_{i,i-1} &= \frac{1}{4} \quad 0 < i < k \\ P_{i,k-i} &= \frac{1}{4} \quad 0 < i < k \\ P_{i,k-i+1} &= \frac{1}{4} \quad 0 < i < k \end{aligned}$$

The first equation refers to the situation where the runner returns to the same door she left from and then chooses that door the next day; the second equation refers to the situation where the runner returns to the opposite door from which she left from and then she chooses the original the next day; and so on. Also we have

$$\begin{aligned} P_{0,0} &= \frac{1}{2} \\ P_{0,k} &= \frac{1}{2} \\ P_{k,k} &= \frac{1}{4} \\ P_{k,0} &= \frac{1}{4} \\ P_{k,1} &= \frac{1}{4} \\ P_{k,k-1} &= \frac{1}{4} \end{aligned}$$

It is also each to check that the Markov chain is also doubly stochastic, so the long run proportions are equal. Hence, the proportion of time the runner runs barefooted is $\frac{1}{k+1}$.

5. This is just Gambler's Ruin Problem (with winning probability p) where the player starts with m units and tries to reach $n+m$ before reaching 0. Thus the desired probability is

$$P_m = \begin{cases} \frac{1-(q/p)^m}{1-(q/p)^{m+n}} & \text{if } p \neq \frac{1}{2} \\ \frac{m}{m+n} & \text{if } p = \frac{1}{2} \end{cases}$$

where $q = 1 - p$ as usual.

6. We know that state $1, 2, 3$ are transient states, so we have

$$\mathbf{P}_T = \begin{bmatrix} 0.4 & 0.2 & 0.1 \\ 0.1 & 0.5 & 0.2 \\ 0.3 & 0.4 & 0.2 \end{bmatrix}$$

and

$$S = (I - \mathbf{P}_T)^{-1} = \begin{bmatrix} 0.6 & -0.2 & -0.1 \\ -0.1 & 0.5 & -0.2 \\ -0.3 & -0.4 & 0.8 \end{bmatrix}^{-1} = \begin{bmatrix} 320/145 & 40/29 & 90/145 \\ 140/145 & 90/29 & 130/145 \\ 190/145 & 60/29 & 280/145 \end{bmatrix}$$

Since $f_{i3} = \frac{s_{i3}-\delta_{i3}}{s_{33}}$, we have $f_{13} = 9/28, f_{23} = 13/28$ and $f_{33} = 27/56$.

7. (a) First we check $\mu = \sum_{i=0}^{\infty} jP_j = \frac{3}{2} > 1$. Then we have

$$\pi_0 = P_0 + \pi_0^2 P_2 \quad \Rightarrow \quad 3\pi_0^2 - 4\pi_0 + 1 = 0 \quad \Rightarrow \quad \pi_0 = \frac{1}{3} \text{ or } \pi_0 = 1 (\text{ rejected })$$

(b) Again we check $\mu = \sum_{i=0}^{\infty} jP_j = 1$. Thus $\pi_0 = 1$.

(c) Finally we check $\mu = \sum_{i=0}^{\infty} jP_j = \frac{3}{2} > 1$. Then we have

$$\begin{aligned}
&\pi_0 = P_0 + \pi_0 P_1 + \pi_0^3 P_3 \\
\Longrightarrow &2\pi_0^3 - 3\pi_0 + 1 = 0 \\
\Longrightarrow &\left(2\pi_0^2 + 2\pi_0 - 1\right)(\pi_0 - 1) = 0 \\
\Longrightarrow &\pi_0 = \frac{\sqrt{3}-1}{2} \text{ or } \pi_0 = \frac{-\sqrt{3}-1}{2} \text{ (rejected) or } \pi_0 = 1 \text{ (rejected)}
\end{aligned}$$